

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1504

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)

Total Marks: 40

Question Count: 4

Duration : 60 Minutes

Code of Conduct

I **G. Ram Charan (192525043)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: **G. Ram Charan**

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks-2 Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book,author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Aim

To develop a cloud-based **Library Book Reservation System** for SIMATS Library using a Cloud Service Provider and demonstrate the **SaaS** model.

Procedure

1. Create a new cloud application.
2. Add fields: Title, Author, Publication, Year, Edition, No. of Copies, Rack No., and Status.
3. Connect the form to a cloud database.
4. Enter book details and save them.
5. Test book reservation and return status updates.

Books, Report									
		book title ▾	book id ▾	author ▾	publication ▾	year ▾	edition ▾	No of copies ▾	
		database management system	1003	Raghu Ramakrishnan	McGraw-Hill	2021	4	8	
		data structures using c	1002	Reema thareja	Oxford University Press	2022	2	10	
		introduction to python programming	1001	John zele	pearson education	2023	3	15	

Students Report									
		Student id ▾	student name ▾	Register Number ▾	Department ▾	Email ▾	Phone Number ▾		
		103	vijay kumar	19252026	IT	vijaykumar26@gmail.com	+918838663398		
		102	ragul kumar	19234568	AIML	ragulkumar89@gmail.com	+919876543212		
		101	aswini sundaram	19253347	CSE	aswini02@gmail.com	+918123456789		

Book Reservation Report									
		Reservation id ▾	Student ▾	book ▾	issue date ▾	return date ▾	status ▾		
		5003	Ms. abinaya	database management system	05-Aug-2026	19-Aug-2026	Returned		
		5002	Mr. rahul	data structures	04-Aug-2026	18-Aug-2026	taken		
		5001	Mrs. aswini	python programming	03-Aug-2026	17-Aug-2026	taken		

Result

A cloud-based Library Book Reservation System was successfully created, enabling online book management and demonstrating the Software as a Service (SaaS) model.

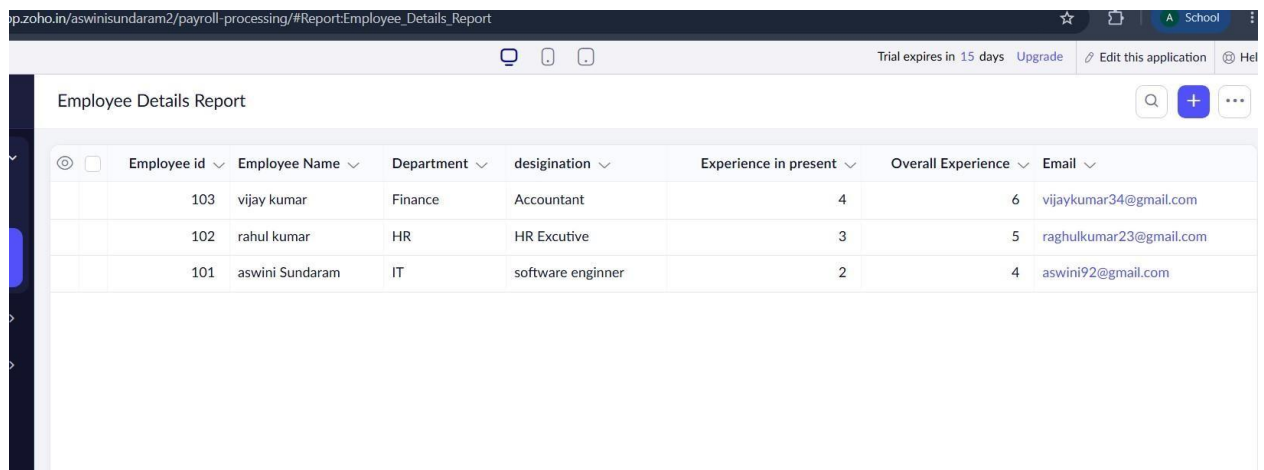
Experiment 2: Create a simple cloud software application for Payroll

Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Aim

To develop a cloud-based **Payroll Processing System** using a Cloud Service Provider to demonstrate the **Software as a Service (SaaS)** model. **Procedure**

1. Create a new cloud application.
2. Add fields: Employee Name, Present Organization Experience, Overall Experience, Basic Pay, HRA, DA, TA, Gross Pay, and Net Pay.
3. Connect the form to a cloud database.
4. Enter employee details and calculate salary.
5. Save and test the payroll records.



The screenshot displays a web application interface for a payroll system. At the top, there's a navigation bar with a search icon, a plus icon, and a menu icon. Below this, the title 'Employee Details Report' is visible. The main content area features a table with the following columns: Employee id, Employee Name, Department, designation, Experience in present, Overall Experience, and Email. The table contains three rows of data:

Employee id	Employee Name	Department	designation	Experience in present	Overall Experience	Email
103	vijay kumar	Finance	Accountant	4	6	vijaykumar34@gmail.com
102	rahul kumar	HR	HR Excutive	3	5	raghulkumar23@gmail.com
101	aswini Sundaram	IT	software enginner	2	4	aswini92@gmail.com

Salary Details Report							Search	+	...
Salary id	Employee Name	Basic Pay	HRA	DA	TA				
203	vijay kumar	10000	8900	9000	7800				
202	rahul kumar	6000	5500	4500	6700				
201	aswini Sundaram	4000	6000	5000	3500				

Payroll Report							Search	+	...
payroll id	Employee	gross pay	deduction	Net pay	pay month				
303	vijay kumar	6587	6474	7686	November				
302	rahul kumar	7465	5747	5263	June				
301	aswini Sundaram	5600	6700	5600	April				

Result

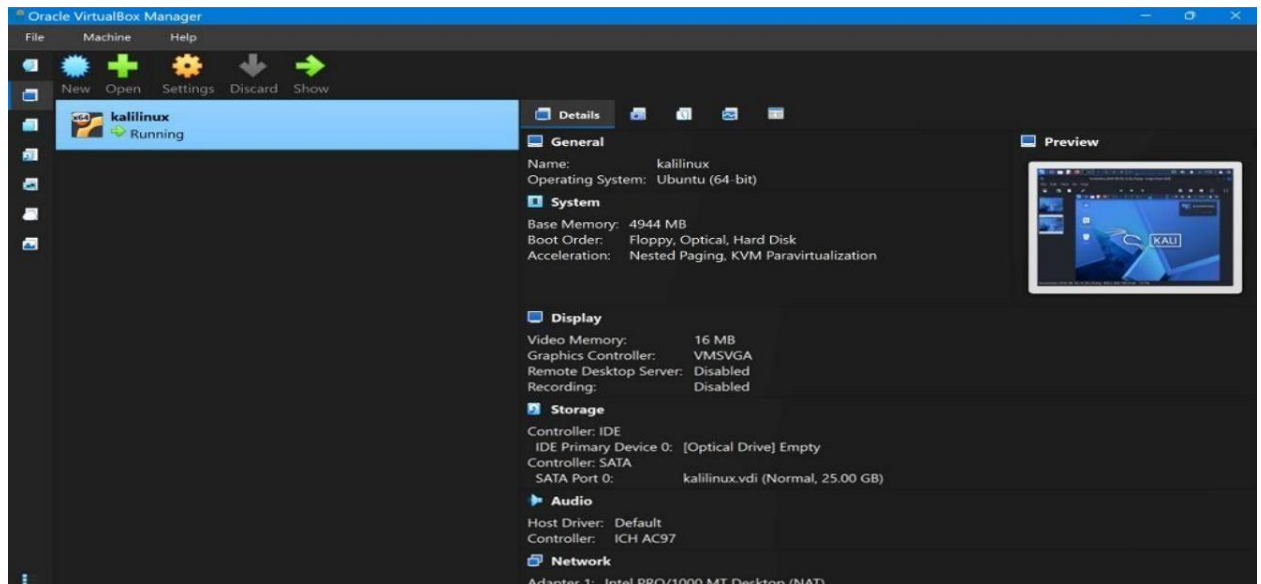
A cloud-based **Payroll Processing System** was successfully created, allowing online payroll management and demonstrating the **Software as a Service (SaaS)** model.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim

To demonstrate virtualization by installing a **Type-2 Hypervisor**, creating a virtual machine with **1 CPU, 2 GB RAM, and 15 GB storage**, and installing a Windows/Linux operating system. **Procedure**

1. Install a Type-2 Hypervisor (e.g., Oracle VirtualBox or VMware Workstation).
2. Create a new Virtual Machine.
3. Allocate **1 CPU, 2 GB RAM, and 15 GB virtual disk**.
4. Install the Windows/Linux operating system using an ISO file.
5. Start the VM and verify that it runs successfully.



Result

A virtual machine was successfully created and configured with **1 CPU, 2 GB RAM, and 15 GB storage**, demonstrating Type-2 virtualization.

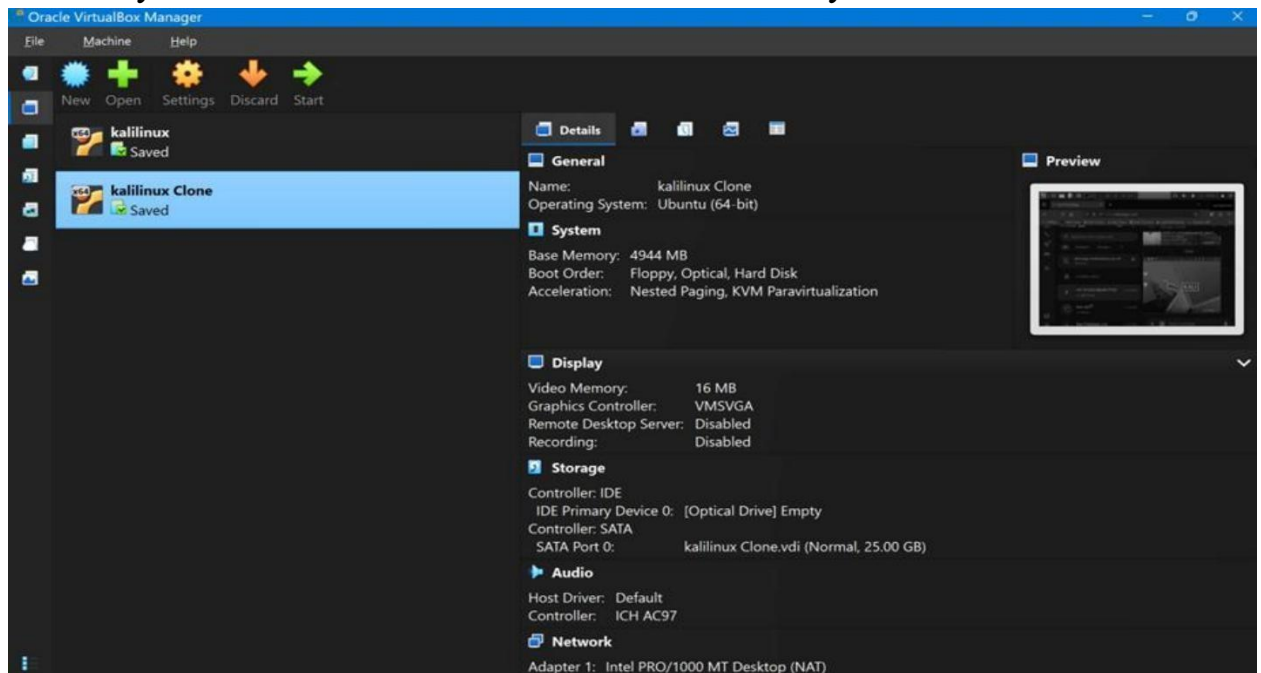
Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Aim

To demonstrate virtualization by creating a **clone of a virtual machine** using a Type-2 Hypervisor and testing the cloned VM.

Procedure

1. Install a Type-2 Hypervisor (VirtualBox/VMware).
2. Create and configure a virtual machine with a Windows/Linux OS.
3. Use the **Clone** option to create a copy of the VM.
4. Start the cloned virtual machine.
5. Verify that the cloned VM loads and works correctly.



Result

A cloned virtual machine was successfully created and tested, demonstrating VM cloning and virtualization using a Type-2 Hypervisor.